

Prerequisite Mapping & Syllabi

Applicant: Riju Marwah — **Target:** M.Sc. CS/AI (Germany, Winter 2026)

Undergrad: B.Tech CSE, GGSIPU — **Language of Instruction:** English

Page 1 provides a one-page prerequisite mapping. The following pages contain concise, unit-level syllabi in the same order, with page references from the table.

Prerequisite Mapping (1-page)

Core Area	Code	Course Title	Sem	ECTS (approx.)	Key Topics (brief)	Ref.
Data Structures & Algorithms	CIC-209; CSE-DAA	Data Structures; Design & Analysis of Algorithms	3,5	~8×1.5	ADTs, trees, graphs, sorting/searching, DP, greedy, complexity	p. 4
Operating Systems	CSE-OS	Operating Systems (+ Lab)	5	~5×1.5	Processes/threads, synchronization, scheduling, memory mgmt, files	p. 5
Computer Networks	CSE-CN	Computer Networks (+ Lab)	5	~5×1.5	TCP/IP, routing, congestion control, transport/app protocols	p. 6
Databases	CSE-DBMS	Database Management Systems (+ Lab)	4	~4×1.5	ER/relational, SQL, normalization, transactions, indexing	p. 7
PL & Compilers	CIC-201; CSE-Java; CSE-AJava; CSE-Comp	Prog. in C; Java; Advanced Java; Compiler Design (+ Lab)	2,4,6,5	~14×1.5	Lexing/parsing, IR/codegen, OOP, JVM, exceptions, I/O	p. 8
Software Eng.	CSE-SE; CSE-SPM	Software Engineering; Software Project Management (+ Labs)	5,6	~6×1.5	SDLC, UML, testing, agile, estimation, risk	p. 9
AI / ML	CSE-AI; ML-407; ML-409; ML-411	AI; Machine Learning; RL & DL; Pattern Recognition & CV	6–7	~12×1.5	Search/logic, SVMs, CNNs, RL, vision, evaluation	p. 10
Web Tech	CSE-WT	Web Technologies (+ Lab)	6	~4×1.5	HTML/CSS/JS, REST/JSON, auth, basic full-stack	p. 11
Theory of Computation	CSE-TOC	Theory of Computation	4	~4×1.5	Automata, CFG/PDA, TM, decidability, complexity	p. 12
Discrete Mathematics	CSE-DM	Discrete Mathematics	3	~4×1.5	Logic, sets, combinatorics, graphs, recurrences	p. 13
Digital Logic & CA	ECC-207	Digital Logic & Computer Design (+ Lab)	3	~4×1.5	Boolean algebra, sequential circuits, CPU/ALU, memory	p. 14
Computational Methods	ES-201	Computational Methods	3	~4×1.5	Root finding, interpolation, numerical integration, ODE/PDE	p. 15
Mathematics (Calc & LA)	AM-I; AM-II	Applied Mathematics I–II	1–2	~8×1.5	Limits, diff/int, matrices, eigenvalues, series	p. 16
Probability & Statistics	PSLP; SSMDA	Prob/Stats & Linear Programming (+ Lab); Statistical Modelling & Analytics	4,6	~8×1.5	RVs, distributions, estimation, tests, regression, LP	p. 17
Foundations & Systems	CIC-201; CIC-203	Programming in C (+ Lab); Electrical Science (+ Lab)	2	~4×1.5	C programming, pointers, memory; DC/AC circuits, machines	p. 18
Entrepreneurship (HS)	PEM	Principles of Entrepreneurship Mindset	7	~2×1.5	Opportunity recognition, problem framing, venture basics	p. 19

Document Structure

- Page 1 — Prerequisite Mapping (this page is referenced from each row)
- Pages 2-... — Detailed syllabi in the same order as the mapping

Labs reinforce the same outcomes as lectures and are grouped with the parent course to keep this concise.

1 Data Structures; Design & Analysis of Algorithms

Semester 3 & 5

Credits 8 (ECTS $\sim 8 \times 1.5$)

Key Topics

- ADTs, arrays, linked lists, stacks, queues, hashing, disjoint sets
- Trees (BST, AVL, heaps), graphs (BFS/DFS, MST, shortest paths)
- Sorting/searching (bubble, insertion, merge, quick), selection
- Algorithm design: divide&conquer, greedy, dynamic programming; amortized analysis

Learning Outcomes

- Implement core data structures; select appropriate algorithms
- Analyze time/space complexity; reason about correctness

2 Operating Systems (+ Laboratory)

Semester 5

Credits 5 (ECTS $\sim 5 \times 1.5$)

Key Topics

- Processes/threads, CPU scheduling, synchronization, deadlocks
- Memory mgmt: paging, segmentation, virtual memory
- File systems, I/O, concurrency primitives; POSIX APIs

Learning Outcomes

- Build concurrent programs and explain OS abstraction tradeoffs
- Diagnose performance issues in scheduling/memory/file subsystems

3 Computer Networks (+ Laboratory)

Semester 5

Credits 5 (ECTS $\sim 5 \times 1.5$)

Key Topics

- Internet architecture, TCP/IP, addressing/subnetting, routing
- Transport services (TCP/UDP), reliability and flow/congestion control
- Application protocols, sockets programming, basic measurement

Learning Outcomes

- Design/test client-server systems; reason about latency/throughput
- Use packet tools to troubleshoot network behavior

4 Database Management Systems (+ Laboratory)

Semester 4

Credits 4 (ECTS $\sim 4 \times 1.5$)

Key Topics

- ER & relational models; SQL (DDL/DML), joins, nested queries
- Normalization; transactions, ACID; indexing; basic query optimization

Learning Outcomes

- Design normalized schemas and write optimized SQL
- Explain concurrency/recovery; evaluate indexes and plans

5 Programming in C; Java; Advanced Java; Compiler Design (+ Lab)

Semester 2,4,6,5

Credits 14 (ECTS $\sim 14 \times 1.5$)

Key Topics

- C fundamentals, pointers, memory mgmt, strings, file I/O
- OOP in Java: classes/objects, inheritance, exceptions, collections
- Advanced Java: I/O, multithreading, networking, GUI/web basics
- Compilers: lexical/syntax analysis (LL/LR), AST/IR, code generation, simple optimizations

Learning Outcomes

- Write idiomatic C/Java; use OOP patterns safely and effectively
- Build/understand compiler components from lexing to codegen

6 Software Engineering; Software Project Management (+ Labs)

Semester 5–6

Credits 6 (ECTS $\sim 6 \times 1.5$)

Key Topics

- SDLC models; requirements; UML/design patterns; verification & testing
- Agile/Scrum; estimation; risk/quality; scheduling; CI/CD basics

Learning Outcomes

- Plan, track, and ship software with quality gates
- Apply testing strategies and manage risks/stakeholders

7 Artificial Intelligence; Machine Learning; Reinforcement Learning & Deep Learning; Pattern Recognition & Computer Vision

Semester 6–7

Credits 12 (ECTS $\sim 12 \times 1.5$)

Key Topics

- AI: search, knowledge representation, logic, planning
- ML: supervised/unsupervised basics; SVMs, trees, ensembles; evaluation
- DL: MLP/CNN basics; optimization; regularization; intro to transfer learning
- RL: value/policy methods; exploration; function approximation
- PR&CV: classical features & classifiers; CNNs for vision; metrics

Learning Outcomes

- Train/evaluate ML/DL models with statistical rigor
- Communicate limitations; apply introductory XAI tooling

8 Web Technologies (+ Laboratory)

Semester 6

Credits 4 (ECTS $\sim 4 \times 1.5$)

Key Topics

- HTML/CSS/JS; REST/JSON; sessions/auth; basic security hygiene
- Client–server apps; simple database-backed web projects

Learning Outcomes

- Build and document small full-stack web apps
- Design clean HTTP APIs and handle auth/state correctly

9 Theory of Computation

Semester 4

Credits 4 (ECTS $\sim 4 \times 1.5$)

Key Topics

- Automata (DFA/NFA), regular languages
- CFGs, pushdown automata; pumping lemmas (intro)
- Turing machines; decidability; introduction to complexity classes

Learning Outcomes

- Classify problems by language/complexity
- Construct simple automata/grammars and reason about decidability

10 Discrete Mathematics

Semester 3

Credits 4 (ECTS $\sim 4 \times 1.5$)

Key Topics

- Logic, sets, relations, functions
- Proof methods, induction; counting & recurrences
- Graph theory basics

Learning Outcomes

- Write rigorous proofs; model problems using graphs/sets
- Apply counting techniques and recurrence relations

11 Digital Logic & Computer Design (+ Laboratory)

Semester 3

Credits 4 (ECTS $\sim 4 \times 1.5$)

Key Topics

- Boolean algebra; combinational circuits; K-maps, multiplexers/decoders
- Sequential circuits: flip-flops, registers, counters; state machines
- Computer organization: CPU datapath/control, ALU; memory hierarchy, I/O

Learning Outcomes

- Design/analyze combinational and sequential logic
- Explain basic CPU organization and memory subsystems

12 Computational Methods

Semester 3

Credits 4 (ECTS $\sim 4 \times 1.5$)

Key Topics

- Errors & rounding; Taylor/Rolle; data representation
- Root finding: bisection, Newton, secant; line search & golden section
- Interpolation: Newton/Gregory; splines; numerical integration (NC/Trapezoid/Simpson)
- Linear systems (Gauss, LU, pivoting); eigen intro; ODE/PDE numerics (Euler, RK, predictor-corrector)

Learning Outcomes

- Implement stable numerical routines in C/C++; assess error/complexity
- Select appropriate methods for roots, interpolation, integration, and ODEs

13 Applied Mathematics I–II (Calculus & Linear Algebra)

Semester 1–2

Credits 8 (ECTS $\sim 8 \times 1.5$)

Key Topics

- Limits, differentiation, integration, series; complex numbers (overview)
- Matrices, vector spaces, eigenvalues/eigenvectors, linear transforms

Learning Outcomes

- Solve calculus problems relevant to optimization
- Apply linear algebra to numerical/ML contexts (least squares, PCA intuition)

14 Probability, Statistics & Linear Programming; Statistical Modelling & Data Analytics (+ Labs)

Semester 4 & 6

Credits 8 (ECTS $\sim 8 \times 1.5$)

Key Topics

- Probability, RVs, common distributions; estimation & CIs
- Hypothesis tests (t/chi-square/ANOVA); regression basics
- Linear programming: simplex/duality (intro); basic analytics tooling

Learning Outcomes

- Design statistically sound experiments and model evaluations
- Interpret p-values, CIs, effect sizes; analyze with regression/LP

15 Programming in C (+ Lab); Electrical Science (+ Lab)

Semester 2

Credits 4 (ECTS $\sim 4 \times 1.5$)

Key Topics

- C language: tokens, operators, control flow; arrays, strings
- Pointers, pointer arithmetic, dynamic memory, functions; file handling
- DC/AC circuits; network theorems; machines (DC/AC), transformers; instruments

Learning Outcomes

- Write memory-safe C using pointers/arrays and modular design
- Explain fundamentals of circuits/machines relevant to computing hardware

16 Principles of Entrepreneurship Mindset

Semester 7

Credits 2 (ECTS $\sim 2 \times 1.5$)

Key Topics

- Opportunity recognition; problem framing; customer discovery
- Basics of venture planning: value prop, MVP, go-to-market, ethics

Learning Outcomes

- Formulate a problem/solution fit and communicate a lean plan
- Collaborate effectively and reflect on societal/ethical considerations